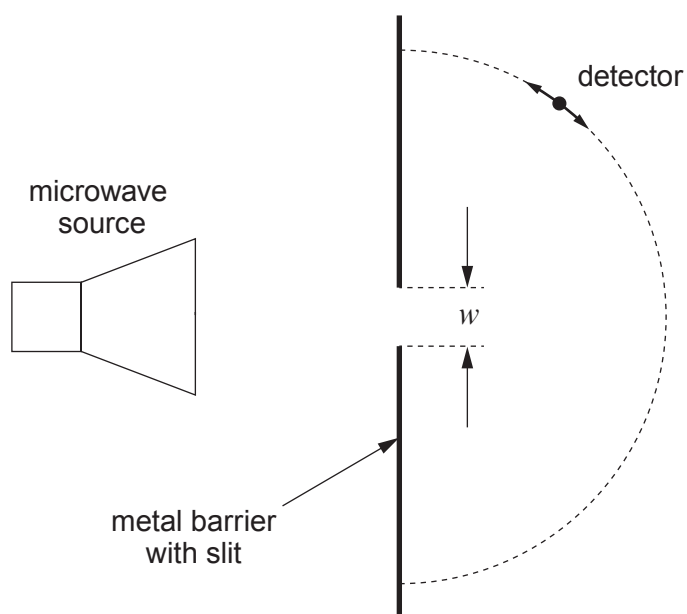


5. (a) In the set-up shown, a microwave detector moved along the semi-circular path detects diffracted waves at **all** points around the semicircle.

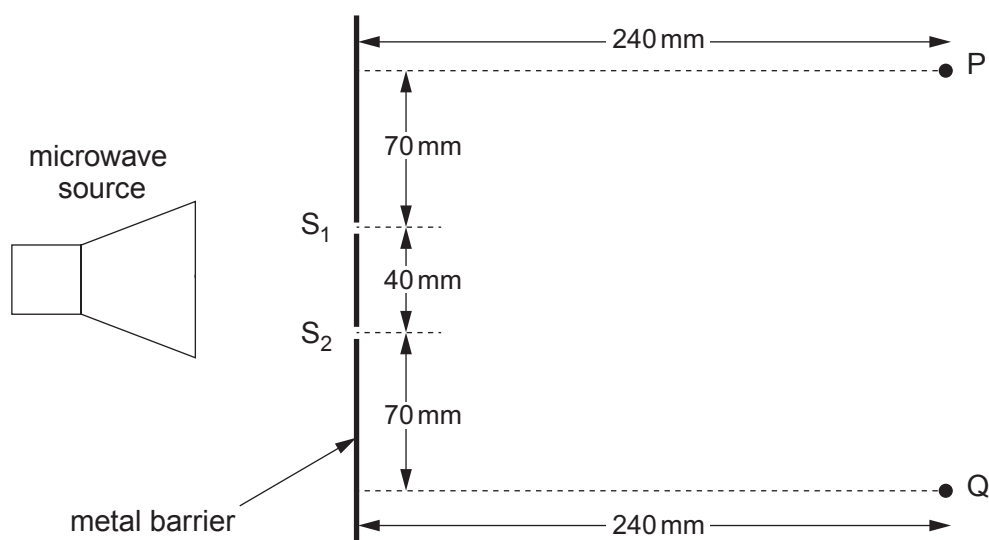


- (i) The width,  $w$ , of the gap is 25 mm. What can be deduced about the wavelength of the microwaves? [1]

- (ii) Rhian suggests that making the gap wider will increase the intensity of the microwaves detected. Discuss to what extent she is correct. [2]



- (b) In the set-up shown, slits  $S_1$  and  $S_2$  act as in-phase sources. Rhian detects **minima** of wave intensity at P and Q, and a single maximum between P and Q.



*Diagram not to scale*

Calculate the path lengths  $S_1P$  and  $S_2P$ , and **hence** the wavelength of the microwaves. [4]

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